```
  ___  ____  ____  ____  ____(R)
 /__    /   ____/   /   ____/
___/   /   /___/   /   /___/
  Statistics/Data Analysis

     User: Jakub Murcek
  Project: Master Thesis
```

```
. do "C:\Users\murcj\Projects\glassdoor-jobs-ai-skills\analysis\stata\ai_skills_analysis.do"

. ******************************************************************************
. * AI SKILLS IN IT JOB POSTINGS - STATA ANALYSIS
. * ============================================================================
. * Dataset: us_relevant_ai.csv
. * Autor: [Yakub Murcek]
. * Datum: Leden 2026
. *
. * Tento do-file obsahuje kompletní analýzu datasetu IT pracovních inzerátů
. * s fokusem na požadavky na AI dovednosti.
. *
. * TESTOVÁNO PRO: STATA 15.1 (IC/SE verze)
. ******************************************************************************
.
. * ============================================================================
. * 1. NASTAVENÍ PROSTŘEDÍ
. * ============================================================================
. * Vyčistíme paměť a nastavíme working directory
. * Toto je důležité pro reprodukovatelnost - každý běh začíná čistě
. * AUTO-SETUP: Pokus o automatické nastavení složky (Windows)
. capture {

. if _rc {
.     display as text "Pozor: Nepodařilo se automaticky nastavit složku. Ujistěte se, že jste v
. }

. display "Aktuální pracovní složka: " c(pwd)
Aktuální pracovní složka: C:\Users\murcj\Projects\glassdoor-jobs-ai-skills\analysis\stata

.
. version 15.1

. clear all

. set more off

.
. * Nastav cestu k datům - UPRAV PODLE SVÉ STRUKTURY
. global datadir "../../data/outputs"

. global outdir "./output"

.
. * Vytvoř output složku pokud neexistuje
. capture mkdir "$outdir"

.
. * Log file - zaznamenává veškerý výstup pro pozdější kontrolu
. capture log close

. * Generovani casoveho razitka pro unikatni nazev log souboru
. local c_date = c(current_date)

. local c_time = c(current_time)

. local time_string = subinstr("`c_date'_`c_time'", " ", "_", .)

. local time_string = subinstr("`time_string'", ":", "-", .)
```

```
45 . log using "$outdir/ai_skills_analysis_`time_string'.log", replace text
   (note: file C:\Users\murcj\Projects\glassdoor-jobs-ai-skills\analysis\stata\output/ai_skills_an
   ------------------------------------------------------------------------------------------------
         name:  <unnamed>
          log:  C:\Users\murcj\Projects\glassdoor-jobs-ai-skills\analysis\stata\output/ai_skills_a
     log type:  text
    opened on:  24 Jan 2026, 13:02:46

46 .
47 .
48 . * ============================================================================
49 . * 2. IMPORT DAT
50 . * ============================================================================
51 . * Používáme soubor 'us_relevant_ai_stata.csv'
52 . import delimited "$datadir/us_relevant_ai_stata.csv", delimiter(";") clear varnames(1) encodi
   (72 vars, 18,464 obs)

53 .
54 . * Základní kontrola úspěšného importu
55 . describe

   Contains data
     obs:        18,464
    vars:            72
    size:    88,479,488
   ------------------------------------------------------------------------------------------------
                 storage   display    value
   variable name   type    format     label      variable label
   ------------------------------------------------------------------------------------------------
   skills          str576  %576s
   skills_ai_det   str98   %98s
   skills_hasai_~t byte    %8.0g
   desc_tier_llm   str14   %14s
   desc_conf_llm   float   %9.0g
   ai_confidence   float   %9.0g
   desc_ai_llm     str400  %400s
   is_real_ai      byte    %8.0g
   hardskills      str934  %934s
   softskills      str452  %452s
   edu_level_det   str10   %10s
   edulevel_llm    str11   %11s
   experience_mi~m float   %9.0g
   ai_det_llm_ma~h byte    %8.0g
   desc_hard_det   str934  %934s
   desc_soft_det   str452  %452s
   desc_hard_llm   byte    %8.0g
   desc_soft_llm   byte    %8.0g
   id              int     %8.0g
   job_title       str131  %131s
   location        str32   %32s
   company_id      long    %12.0g
   company         str81   %81s
   age_in_days     int     %8.0g
   pay_currency    str3    %9s
   pay_period      str7    %9s
   salary_min      float   %9.0g
   salary_mid      float   %9.0g
   salary_max      float   %9.0g
   rating          float   %9.0g
   discover_date   str19   %19s
   job_types       str73   %73s
   remote_work_t~s str14   %14s
   city            str32   %32s
   state           str24   %24s
   country         str13   %13s
   latitude        float   %9.0g
   longitude       float   %9.0g
   ceo             str64   %64s
   headquarters    str37   %37s
   industry        str40   %40s
   sector          str43   %43s
   revenue         str32   %32s
```

```
size            str23   %23s
type            str30   %30s
website         str127  %127s
year_founded    int     %8.0g
cluster_archi~s byte    %8.0g
cluster_bi___~s byte    %8.0g
cluster_backe~t byte    %8.0g
cluster_certi~s byte    %8.0g
cluster_cloud~g byte    %8.0g
cluster_data_~s byte    %8.0g
cluster_data_~g byte    %8.0g
cluster_data_~l byte    %8.0g
cluster_datab~e byte    %8.0g
cluster_devop~s byte    %8.0g
cluster_dynam~b byte    %8.0g
cluster_enter~d byte    %8.0g
cluster_enter~s byte    %8.0g
cluster_front~t byte    %8.0g
cluster_gener~i byte    %8.0g
cluster_legac~e byte    %8.0g
cluster_mobil~p byte    %8.0g
cluster_netwo~g byte    %8.0g
cluster_os___~d byte    %8.0g
cluster_scrip~l byte    %8.0g
cluster_secur~y byte    %8.0g
cluster_syste~g byte    %8.0g
cluster_testi~g byte    %8.0g
cluster_tools~s byte    %8.0g
education_hyb~d str11   %11s
-----------------------------------------------------------------------------------------
Sorted by:
     Note: Dataset has changed since last saved.

56 . display "Počet pozorování (job postů): " _N
   Počet pozorování (job postů): 18464

57 .
58 .
59 . * ============================================================================
60 . * 3. ČIŠTĚNÍ A PŘÍPRAVA DAT
61 . * ============================================================================
62 .
63 . * --- 3.0 Filtrování (podle požadavku uživatele) ---
64 . * Ponecháme pouze pozice s vysokou confidencí modelu
65 . keep if desc_conf_llm >= 0.7
   (607 observations deleted)

66 . display "Počet pozorování po filtrování confidence >= 0.7: " _N
   Počet pozorování po filtrování confidence >= 0.7: 17857

67 .
68 . * --- 3.1 AI Tier klasifikace ---
69 . * desc_tier_llm obsahuje kategorie: "none", "ai_integration", "applied_ai"
70 . replace desc_tier_llm = "missing" if desc_tier_llm == ""
   (0 real changes made)

71 . encode desc_tier_llm, generate(ai_tier_num)

72 .
73 . * --- 3.2 AI Flag (Strict Intersection + Buzzword Filter) ---
```

```
 74 . * 1. Sloučime zdroje skills do jednoho retezce pro kontrolu
 75 . gen skills_combined = lower(skills_ai_det + " " + desc_ai_llm)

 76 .
 77 . * 2. Odstranime obecne buzzwords (AI, ML, Artificial Intelligence atd.)
 78 . * Pouzivame regex s word boundaries \b aby se smazalo jen "AI" a ne "OpenAI"
 79 . gen skills_no_buzz = ustrregexra(skills_combined, "(?i)\b(ai|ml|artificial intelligence|machi

 80 .
 81 . * 3. Odstranime interpunkci (carky, mezery), abychom zarucili ze zbyva neco realneho
 82 . gen skills_cleaned = subinstr(skills_no_buzz, ",", "", .)

 83 . replace skills_cleaned = subinstr(skills_cleaned, " ", "", .)
    (17,857 real changes made)

 84 . replace skills_cleaned = subinstr(skills_cleaned, ";", "", .)
    (0 real changes made)

 85 .
 86 . * 4. Definice has_ai_flag
 87 . * Tier != None A ZAROVEN zbyly nejake specificke skills (delka > 1 znaku)
 88 . gen has_ai_flag = ((desc_tier_llm != "none" & desc_tier_llm != "missing") & length(skills_cle

 89 .
 90 . label variable has_ai_flag "AI Job (Tier + Valid Skills, no buzzwords)"

 91 .
 92 . * Kompatibilita pro starší skripty (volitelné)
 93 . gen has_ai = has_ai_flag

 94 .
 95 . * --- 3.3 Vzdělání (Hybridní) ---
 96 . * Používáme předpřipravenou 'education_hybrid' proměnnou
 97 . * Normalizované hodnoty: "bachelor", "master", "highschool", "missing"
 98 . replace education_hybrid = "highschool" if education_hybrid == "high school"
    (2 real changes made)

 99 . encode education_hybrid, generate(edu_cat)

100 . label variable edu_cat "Pozadovane vzdelani (kategoricke)"

101 .
102 . * --- 3.4 Zkušenosti ---
103 . * experience_min_llm by měl být již numerický, ale pro jistotu
104 . destring experience_min_llm, replace force
    experience_min_llm already numeric; no replace

105 . label variable experience_min_llm "Min. pozadovane roky zkusenosti"

106 .
107 . * Kategorie zkušeností
108 . gen exp_category = .
    (17,857 missing values generated)

109 . replace exp_category = 1 if experience_min_llm == 0 | experience_min_llm == .
    (3,348 real changes made)

110 . replace exp_category = 2 if experience_min_llm > 0 & experience_min_llm <= 2
    (2,460 real changes made)
```

```
111 . replace exp_category = 3 if experience_min_llm > 2 & experience_min_llm <= 5
    (8,411 real changes made)

112 . replace exp_category = 4 if experience_min_llm > 5 & experience_min_llm <= 10
    (3,419 real changes made)

113 . replace exp_category = 5 if experience_min_llm > 10 & experience_min_llm < .
    (219 real changes made)

114 .
115 . label define exp_lbl 1 "Entry (0)" 2 "Junior (1-2)" 3 "Mid (3-5)" 4 "Senior (6-10)" 5 "Expert

116 . label values exp_category exp_lbl

117 . label variable exp_category "Kategorie seniority"

118 .
119 . * --- 3.5 Plat ---
120 . destring salary_min salary_mid salary_max, replace force
    salary_min already numeric; no replace
    salary_mid already numeric; no replace
    salary_max already numeric; no replace

121 . replace salary_mid = . if salary_mid < 20000 | salary_mid > 500000
    (908 real changes made, 908 to missing)

122 . label variable salary_mid "Rocni plat - stredni hodnota (USD)"

123 .
124 . * --- 3.6 Sektor a Lokace ---
125 . replace sector = "Unknown" if sector == ""
    (3,504 real changes made)

126 . replace industry = "Unknown" if industry == ""
    (3,505 real changes made)

127 . encode sector, generate(sector_num)

128 .
129 . replace state = "Unknown" if state == ""
    (4,549 real changes made)

130 . encode state, generate(state_num)

131 .
132 . * 3.7 Remote práce ---
133 . * Zde odstranime "capture confirm" abychom videli jestli promenna existuje
134 . gen is_remote = 0

135 . replace is_remote = 1 if strpos(lower(remote_work_types), "home") > 0
    (5,070 real changes made)

136 . replace is_remote = 1 if strpos(lower(remote_work_types), "remote") > 0
    (0 real changes made)

137 .
138 . label variable is_remote "Moznost remote prace (1=ano)"

139 .
140 .
```

```
141 .
142 . * ==============================================================================
143 . * 4. DESKRIPTIVNÍ STATISTIKA
144 . * ==============================================================================
145 . * Základní přehled datasetu - toto jde typicky do první tabulky v diplomce
146 .
147 . display _n "============================================================="

    ==============================================================

148 . display "4. DESKRIPTIVNI STATISTIKA"
    4. DESKRIPTIVNI STATISTIKA

149 . display "============================================================="
    ==============================================================

150 .
151 . * --- 4.1 Frekvence AI tier klasifikace ---
152 . * KLÍČOVÁ TABULKA: Kolik % pozic vyžaduje AI dovednosti?
153 . display _n "--- 4.1 Distribuce AI pozadavku v IT pozicich ---"

    --- 4.1 Distribuce AI pozadavku v IT pozicich ---

154 . tab desc_tier_llm, missing

     desc_tier_llm |      Freq.     Percent        Cum.
    ---------------+-----------------------------------
    ai_integration |      2,354       13.18       13.18
        applied_ai |      1,307        7.32       20.50
           core_ai |          6        0.03       20.54
              none |     14,190       79.46      100.00
    ---------------+-----------------------------------
             Total |     17,857      100.00

155 . tab desc_tier_llm if desc_tier_llm != "missing", sort

     desc_tier_llm |      Freq.     Percent        Cum.
    ---------------+-----------------------------------
              none |     14,190       79.46       79.46
    ai_integration |      2,354       13.18       92.65
        applied_ai |      1,307        7.32       99.97
           core_ai |          6        0.03      100.00
    ---------------+-----------------------------------
             Total |     17,857      100.00

156 .
157 . * Procentuální rozdělení (pro text diplomky)
158 . count if has_ai == 1
      3,477

159 . local n_ai = r(N)

160 . count
      17,857

161 . local n_total = r(N)

162 . display _n "Podil pozic s AI pozadavky: " %5.2f (`n_ai'/`n_total')*100 "%"

    Podil pozic s AI pozadavky: 19.47%
```

```
163 .
164 . * --- 4.2 Požadované vzdělání ---
165 . display _n "--- 4.2 Distribuce vzdelavacich pozadavku ---"
```

--- 4.2 Distribuce vzdelavacich pozadavku ---

```
166 . tab edu_cat, missing
```

Pozadovane vzdelani (kategoricke)	Freq.	Percent	Cum.
associate	356	1.99	1.99
bachelor	10,357	58.00	59.99
highschool	411	2.30	62.29
master	352	1.97	64.27
missing	6,368	35.66	99.93
phd	13	0.07	100.00
Total	17,857	100.00	

```
167 . tab edu_cat if edu_cat > 0
```

Pozadovane vzdelani (kategoricke)	Freq.	Percent	Cum.
associate	356	1.99	1.99
bachelor	10,357	58.00	59.99
highschool	411	2.30	62.29
master	352	1.97	64.27
missing	6,368	35.66	99.93
phd	13	0.07	100.00
Total	17,857	100.00	

```
168 .
169 . * Porovnání vzdělání podle AI tier
170 . display _n "Vzdelani x AI tier (kontingencni tabulka):"
```

Vzdelani x AI tier (kontingencni tabulka):

```
171 . tab edu_cat ai_tier_num if edu_cat > 0, chi2 column
```

Key
frequency *column percentage*

Pozadovane vzdelani (kategoricke)	ai_tier_num				
	ai_integr	applied_a	core_ai	none	Total
associate	20 0.85	9 0.69	0 0.00	327 2.30	356 1.99
bachelor	1,214 51.57	702 53.71	1 16.67	8,440 59.48	10,357 58.00
highschool	28 1.19	20 1.53	0 0.00	363 2.56	411 2.30
master	26 1.10	39 2.98	2 33.33	285 2.01	352 1.97
missing	1,066 45.28	532 40.70	2 33.33	4,768 33.60	6,368 35.66

phd	0 0.00	5 0.38	1 16.67	7 0.05	13 0.07
Total	2,354 100.00	1,307 100.00	6 100.00	14,190 100.00	17,857 100.00

Pearson chi2(**15**) = **461.9818**   Pr = **0.000**

```
172 .
173 . * --- 4.3 Požadované zkušenosti ---
174 . display _n "--- 4.3 Distribuce pozadavku na zkusenosti ---"
```

--- 4.3 Distribuce pozadavku na zkusenosti ---

```
175 . summarize experience_min_llm, detail
```

Min. pozadovane roky zkusenosti

	Percentiles	Smallest		
1%	0	0		
5%	.5	0		
10%	1	0	Obs	15,262
25%	3	0	Sum of Wgt.	15,262
50%	5		Mean	4.520393
		Largest	Std. Dev.	2.62528
75%	5	20		
90%	8	25	Variance	6.892093
95%	10	25	Skewness	.9727604
99%	12	35	Kurtosis	6.38147

```
176 .
177 . tab exp_category, missing
```

Kategorie seniority	Freq.	Percent	Cum.
Entry (0)	3,348	18.75	18.75
Junior (1-2)	2,460	13.78	32.53
Mid (3-5)	8,411	47.10	79.63
Senior (6-10)	3,419	19.15	98.77
Expert (10+)	219	1.23	100.00
Total	17,857	100.00	

```
178 . tab exp_category ai_tier_num, chi2 column
```

Key
frequency *column percentage*

Kategorie	ai_tier_num				
seniority	ai_integr	applied_a	core_ai	none	Total
Entry (0)	341 14.49	222 16.99	1 16.67	2,784 19.62	3,348 18.75
Junior (1-2)	276 11.72	153 11.71	1 16.67	2,030 14.31	2,460 13.78
Mid (3-5)	1,238 52.59	618 47.28	2 33.33	6,553 46.18	8,411 47.10
Senior (6-10)	472 20.05	289 22.11	2 33.33	2,656 18.72	3,419 19.15
Expert (10+)	27 1.15	25 1.91	0 0.00	167 1.18	219 1.23

Total	2,354 100.00	1,307 100.00	6 100.00	14,190 100.00	17,857 100.00

Pearson chi2(**12**) = **77.2871** Pr = **0.000**

```
179 .
180 . * --- 4.4 Platy ---
181 . display _n "--- 4.4 Distribuce platu ---"
```

--- 4.4 Distribuce platu ---

```
182 . summarize salary_mid, detail
```

Rocni plat - stredni hodnota (USD)

	Percentiles	Smallest		
1%	56538	20000		
5%	70937	30000		
10%	79863	30000	Obs	13,743
25%	95000	32640	Sum of Wgt.	13,743
50%	118480		Mean	125240.4
		Largest	Std. Dev.	42210
75%	148750	445000		
90%	180000	445000	Variance	1.78e+09
95%	200000	445000	Skewness	1.370318
99%	245000	450000	Kurtosis	7.794587

```
183 .
184 . * Platy podle AI tier - DŮLEŽITÉ pro argument o "AI premium"
185 . display _n "Plat podle AI tier:"
```

Plat podle AI tier:

```
186 . tabstat salary_mid, by(desc_tier_llm) statistics(count mean sd min p25 p50 p75 max)
```

Summary for variables: salary_mid
by categories of: desc_tier_llm

desc_tier_llm	N	mean	sd	min	p25	p50	p75	ma
ai_integration	1869	141477.4	46393.67	36500	107987	135000	169600	41000
applied_ai	1041	151004.5	48527	46532	116373	149500	180000	44500
core_ai	4	168045.8	55785.28	100558	129279	167937.5	206812.5	23575
none	10829	119945.5	39048.28	20000	92500	114067	140000	45000
Total	13743	125240.4	42210	20000	95000	118480	148750	45000

```
187 .
188 . * --- 4.5 Sektory a industrie ---
189 . display _n "--- 4.5 Top 10 sektoru ---"
```

--- 4.5 Top 10 sektoru ---

```
190 . tab sector if sector != "Unknown", sort
```

sector	Freq.	Percent	Cum.
Information Technology	6,074	42.32	42.32
Manufacturing	1,219	8.49	50.81
Financial Services	1,081	7.53	58.34
Aerospace & Defense	981	6.83	65.18
Management & Consulting	707	4.93	70.10
Healthcare	692	4.82	74.93
Media & Communication	546	3.80	78.73
Retail & Wholesale	405	2.82	81.55
Education	375	2.61	84.16
Insurance	321	2.24	86.40
Human Resources & Staffing	294	2.05	88.45
Government & Public Administration	271	1.89	90.34

Pharmaceutical & Biotechnology	229	1.60	91.93
Energy, Mining & Utilities	184	1.28	93.21
Telecommunications	175	1.22	94.43
Construction, Repair & Maintenance Se..	171	1.19	95.62
Transportation & Logistics	143	1.00	96.62
Arts, Entertainment & Recreation	103	0.72	97.34
Real Estate	94	0.65	97.99
Nonprofit & NGO	76	0.53	98.52
Personal Consumer Services	56	0.39	98.91
Restaurants & Food Service	48	0.33	99.25
Legal	44	0.31	99.55
Hotels & Travel Accommodation	33	0.23	99.78
Agriculture	30	0.21	99.99
Media	1	0.01	100.00
Total	14,353	100.00	

```
191 .
192 . display _n "--- 4.6 Remote prace ---"
```

--- 4.6 Remote prace ---

```
193 . tab is_remote
```

Moznost remote prace (1=ano)	Freq.	Percent	Cum.
0	12,787	71.61	71.61
1	5,070	28.39	100.00
Total	17,857	100.00	

```
194 . tab is_remote has_ai, chi2 row
```

Key
frequency *row percentage*

Moznost remote prace (1=ano)	has_ai 0	1	Total
0	10,618 83.04	2,169 16.96	12,787 100.00
1	3,762 74.20	1,308 25.80	5,070 100.00
Total	14,380 80.53	3,477 19.47	17,857 100.00

Pearson chi2(**1**) = **180.7841**   Pr = **0.000**

```
195 .
```

```
196 .
197 . * ==========================================================================
198 . * 5. ANALYTICKÉ TESTY - HYPOTÉZY
199 . * ==========================================================================
200 . * Statistické testy pro ověření hypotéz diplomové práce
201 .
202 . display _n "============================================================="

    ==============================================================

203 . display "5. STATISTICKE TESTY"
    5. STATISTICKE TESTY

204 . display "============================================================="
    ==============================================================

205 .
206 . * --- 5.1 T-test: Liší se platy AI vs non-AI pozic? ---
207 . * H0: Průměrný plat AI pozic = průměrný plat non-AI pozic
208 . * H1: Průměrné platy se liší
209 .
210 . display _n "--- 5.1 T-test: Plat AI vs non-AI pozic ---"

    --- 5.1 T-test: Plat AI vs non-AI pozic ---

211 . ttest salary_mid, by(has_ai)
```

Two-sample t test with equal variances

Group	Obs	Mean	Std. Err.	Std. Dev.	[95% Conf.	Interval]
0	10,988	120345.9	375.8993	39403.15	119609.1	121082.8
1	2,755	144761.4	897.5534	47110.87	143001.5	146521.4
combined	13,743	125240.4	360.0597	42210	124534.6	125946.2
diff		-24415.49	874.948		-26130.5	-22700.47

```
    diff = mean(0) - mean(1)                                     t = -27.9051
Ho: diff = 0                                    degrees of freedom =    13741

    Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
 Pr(T < t) = 0.0000         Pr(|T| > |t|) = 0.0000          Pr(T > t) = 1.0000

212 .
213 . * Efekt size (Cohenovo d) - důležité pro interpretaci praktické významnosti
214 . quietly summarize salary_mid if has_ai == 0

215 . local mean_no_ai = r(mean)

216 . local sd_no_ai = r(sd)

217 . local n_no_ai = r(N)

218 .
219 . quietly summarize salary_mid if has_ai == 1

220 . local mean_ai = r(mean)

221 . local sd_ai = r(sd)
```

```
. local n_ai = r(N)

.
. * Pooled standard deviation
. local sd_pooled = sqrt(((`n_no_ai'-1)*`sd_no_ai'^2 + (`n_ai'-1)*`sd_ai'^2) / (`n_no_ai'+`n_ai

. local cohens_d = (`mean_ai' - `mean_no_ai') / `sd_pooled'

.
. display _n "Cohenovo d (effect size): " %5.3f `cohens_d'
```

Cohenovo d (effect size): 0.595

```
. display "Interpretace: |d| < 0.2 = maly, 0.2-0.8 = stredni, > 0.8 = velky efekt"
```

Interpretace: |d| < 0.2 = maly, 0.2-0.8 = stredni, > 0.8 = velky efekt

```
.
. * --- 5.2 ANOVA: Liší se platy mezi AI tiers? ---
. * Testuje rozdíly mezi none, ai_integration, ai_focused
.
. display _n "--- 5.2 ANOVA: Plat podle AI tier ---"
```

--- 5.2 ANOVA: Plat podle AI tier ---

```
. oneway salary_mid ai_tier_num, tabulate bonferroni
```

	Summary of Rocni plat - stredni hodnota (USD)		
ai_tier_num	Mean	Std. Dev.	Freq.
ai_integr	**141477.39**	**46393.668**	**1,869**
applied_a	**151004.53**	**48526.999**	**1,041**
core_ai	**168045.75**	**55785.282**	**4**
none	**119945.51**	**39048.279**	**10,829**
Total	**125240.41**	**42209.997**	**13,743**

Analysis of Variance

Source	SS	df	MS	F	Prob > F
Between groups	**1.4947e+12**	**3**	**4.9823e+11**	**297.75**	**0.0000**
Within groups	**2.2989e+13**	**13739**	**1.6733e+09**		
Total	**2.4484e+13**	**13742**	**1.7817e+09**		

Bartlett's test for equal variances: chi2(**3**) = **172.4292** Prob>chi2 = **0.000**

Comparison of Rocni plat - stredni hodnota (USD) by ai_tier_num
(Bonferroni)

Row Mean-Col Mean	ai_integ	applied_	core_ai
applied_	**9527.14** **0.000**		
core_ai	**26568.4** **1.000**	**17041.2** **1.000**	
none	**-21531.9** **0.000**	**-31059** **0.000**	**-48100.2** **0.112**

```
236 .
237 . * --- 5.3 Chi-square: Vzdělání a AI požadavky ---
238 . * Jsou AI pozice náročnější na vzdělání?
239 .
240 . display _n "--- 5.3 Chi-square: Vzdelani x AI tier ---"
```

--- 5.3 Chi-square: Vzdelani x AI tier ---

```
241 . tab edu_cat has_ai if edu_cat > 0, chi2 expected
```

Key
frequency   *expected frequency*

Pozadovane vzdelani (kategoricke)	has_ai 0	1	Total
associate	**328** **286.7**	**28** **69.3**	**356** **356.0**
bachelor	**8,543** **8,340.4**	**1,814** **2,016.6**	**10,357** **10,357.0**
highschool	**365** **331.0**	**46** **80.0**	**411** **411.0**
master	**286** **283.5**	**66** **68.5**	**352** **352.0**
missing	**4,851** **5,128.1**	**1,517** **1,239.9**	**6,368** **6,368.0**
phd	**7** **10.5**	**6** **2.5**	**13** **13.0**
Total	**14,380** **14,380.0**	**3,477** **3,477.0**	**17,857** **17,857.0**

Pearson chi2(**5**) = **156.7363**   Pr = **0.000**

```
242 .
243 . * --- 5.4 Chi-square: Zkušenosti a AI požadavky ---
244 . display _n "--- 5.4 Chi-square: Zkusenosti x AI tier ---"
```

--- 5.4 Chi-square: Zkusenosti x AI tier ---

```
245 . tab exp_category has_ai, chi2 expected
```

Key
frequency   *expected frequency*

Kategorie seniority	has_ai 0	1	Total
Entry (0)	**2,809** **2,696.1**	**539** **651.9**	**3,348** **3,348.0**
Junior (1-2)	**2,052** **1,981.0**	**408** **479.0**	**2,460** **2,460.0**
Mid (3-5)	**6,655** **6,773.3**	**1,756** **1,637.7**	**8,411** **8,411.0**

Senior (6-10)	2,693 2,753.3	726 665.7	3,419 3,419.0
Expert (10+)	171 176.4	48 42.6	219 219.0
Total	14,380 14,380.0	3,477 3,477.0	17,857 17,857.0

```
          Pearson chi2(4) =  55.5656   Pr = 0.000

. 
. * --- 5.5 Mann-Whitney U test (neparametrický) ---
. * Pro případ, že platy nemají normální rozdělení
. 
. display _n "--- 5.5 Mann-Whitney U test: Plat AI vs non-AI ---"

--- 5.5 Mann-Whitney U test: Plat AI vs non-AI ---

. ranksum salary_mid, by(has_ai)

Two-sample Wilcoxon rank-sum (Mann-Whitney) test
```

has_ai	obs	rank sum	expected
0	10988	70557704	75509536
1	2755	23884192	18932360
combined	13743	94441896	94441896

```
unadjusted variance   3.467e+10
adjustment for ties  -595872.38
                     ----------
adjusted variance     3.467e+10

Ho: salary~d(has_ai==0) = salary~d(has_ai==1)
             z = -26.594
    Prob > |z| =   0.0000

. 
. 
. * ============================================================================
. * 6. REGRESNÍ ANALÝZA
. * ============================================================================
. * Modelování faktorů ovlivňujících plat a AI požadavky
. 
. display _n "=============================================================="

==============================================================

. display "6. REGRESNI ANALYZA"
6. REGRESNI ANALYZA

. display "=============================================================="
==============================================================

. 
. * --- 6.1 OLS regrese: Co ovlivňuje plat? ---
. * Závislá proměnná: salary_mid
. * Nezávislé: has_ai, edu_level, experience_min_llm, is_remote
```

```
.
. display _n "--- 6.1 OLS regrese: Determinanty platu ---"
```

--- 6.1 OLS regrese: Determinanty platu ---

```
.
. regress salary_mid has_ai i.edu_cat experience_min_llm is_remote
```

Source	SS	df	MS
Model	4.3534e+12	8	5.4417e+11
Residual	1.6075e+13	11,890	1.3520e+09
Total	2.0428e+13	11,898	1.7169e+09

Number of obs	=	11,899
F(8, 11890)	=	402.51
Prob > F	=	0.0000
R-squared	=	0.2131
Adj R-squared	=	0.2126
Root MSE	=	36769

salary_mid	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
has_ai	20579.14	843.7757	24.39	0.000	18925.2	22233.08
edu_cat						
bachelor	11716.94	2323.93	5.04	0.000	7161.657	16272.22
highschool	6016.042	3080.548	1.95	0.051	-22.33649	12054.42
master	18683.89	3305.366	5.65	0.000	12204.83	25162.95
missing	18347.54	2377.823	7.72	0.000	13686.62	23008.46
phd	38267.87	13215.18	2.90	0.004	12363.96	64171.77
experience_min_llm	5454.117	132.3355	41.21	0.000	5194.718	5713.516
is_remote	11611.15	813.6178	14.27	0.000	10016.33	13205.98
_cons	80959.43	2312.309	35.01	0.000	76426.92	85491.93

```
.
. * Robustní standardní chyby (heteroskedasticita)
. display _n "--- 6.1b OLS s robustnimi standardnimi chybami ---"
```

--- 6.1b OLS s robustnimi standardnimi chybami ---

```
. regress salary_mid has_ai i.edu_cat experience_min_llm is_remote, vce(robust)
```

Linear regression

Number of obs	=	11,899
F(8, 11890)	=	329.88
Prob > F	=	0.0000
R-squared	=	0.2131
Root MSE	=	36769

salary_mid	Coef.	Robust Std. Err.	t	P>|t|	[95% Conf.	Interval]
has_ai	20579.14	936.2877	21.98	0.000	18743.87	22414.42
edu_cat						
bachelor	11716.94	1999.883	5.86	0.000	7796.841	15637.04
highschool	6016.042	2547.348	2.36	0.018	1022.823	11009.26
master	18683.89	2979.01	6.27	0.000	12844.54	24523.23
missing	18347.54	2112.424	8.69	0.000	14206.84	22488.23
phd	38267.87	14573.83	2.63	0.009	9700.774	66834.96
experience_min_llm	5454.117	158.8186	34.34	0.000	5142.807	5765.428
is_remote	11611.15	853.5739	13.60	0.000	9938.009	13284.3
_cons	80959.43	1980.567	40.88	0.000	77077.19	84841.66

```
274 .
275 . * --- 6.2 Rozšířený model s interakcemi ---
276 . * AI × vzdělání interakce - má AI větší vliv u vyššího vzdělání?
277 .
278 . display _n "--- 6.2 Model s interakcemi ---"
```

--- 6.2 Model s interakcemi ---

```
279 . regress salary_mid c.has_ai##i.edu_cat experience_min_llm is_remote, vce(robust)
```

```
Linear regression                               Number of obs     =     11,899
                                                F(13, 11885)      =     207.30
                                                Prob > F          =     0.0000
                                                R-squared         =     0.2155
                                                Root MSE          =      36720
```

salary_mid	Coef.	Robust Std. Err.	t	P>|t|	[95% Conf.	Interval]
has_ai	19410.84	8892.01	2.18	0.029	1981.044	36840.63
edu_cat						
bachelor	12236.5	2020.524	6.06	0.000	8275.942	16197.06
highschool	7831.508	2607.258	3.00	0.003	2720.855	12942.16
master	17074.39	2952.829	5.78	0.000	11286.36	22862.42
missing	16571.38	2153.314	7.70	0.000	12350.53	20792.23
phd	34644.42	7163.876	4.84	0.000	20602.05	48686.79
edu_cat#c.has_ai						
bachelor	-2364.461	8962.538	-0.26	0.792	-19932.5	15203.58
highschool	-14773.18	10217.54	-1.45	0.148	-34801.24	5254.876
master	8144.445	11337.29	0.72	0.473	-14078.49	30367.38
missing	7096.398	9060.349	0.78	0.434	-10663.37	24856.16
phd	8178.067	30108.66	0.27	0.786	-50839.83	67195.97
experience_min_llm	5458.089	158.7991	34.37	0.000	5146.817	5769.362
is_remote	11639.64	853.8972	13.63	0.000	9965.857	13313.41
_cons	81045.14	1997.531	40.57	0.000	77129.65	84960.62

```
280 .
281 . * --- 6.3 Logistická regrese: Co predikuje AI požadavky? ---
282 . * Které charakteristiky pozice souvisí s AI požadavky?
283 .
284 . display _n "--- 6.3 Logisticka regrese: Prediktory AI pozadavku ---"
```

--- 6.3 Logisticka regrese: Prediktory AI pozadavku ---

```
285 . logit has_ai i.edu_cat experience_min_llm is_remote, or

Iteration 0:   log likelihood = -7640.7448
Iteration 1:   log likelihood = -7497.5094
Iteration 2:   log likelihood = -7494.1649
Iteration 3:   log likelihood = -7494.1444
Iteration 4:   log likelihood = -7494.1444

Logistic regression                             Number of obs     =     15,262
                                                LR chi2(7)        =     293.20
                                                Prob > chi2       =     0.0000
Log likelihood = -7494.1444                     Pseudo R2         =     0.0192
```

has_ai	Odds Ratio	Std. Err.	z	P>|z|	[95% Conf.	Interval]
edu_cat						
bachelor	**2.417142**	**.4997532**	**4.27**	**0.000**	**1.611807**	**3.62486**
highschool	**1.42392**	**.3719776**	**1.35**	**0.176**	**.8533399**	**2.376015**
master	**2.956445**	**.747325**	**4.29**	**0.000**	**1.801377**	**4.85216**
missing	**3.63699**	**.755776**	**6.21**	**0.000**	**2.42025**	**5.465426**
phd	**8.707554**	**6.134277**	**3.07**	**0.002**	**2.18899**	**34.63767**
experience_min_llm	**1.018785**	**.0079704**	**2.38**	**0.017**	**1.003283**	**1.034527**
is_remote	**1.516053**	**.0662321**	**9.52**	**0.000**	**1.391643**	**1.651585**
_cons	**.073278**	**.0151196**	**-12.67**	**0.000**	**.0489039**	**.1098005**

Note: **_cons** estimates baseline odds.

```
286 .
287 . * Marginální efekty (interpretovatelné jako procentní změny)
288 . display _n "--- 6.3b Marginalni efekty ---"
```

--- 6.3b Marginalni efekty ---

```
289 . margins, dydx(*) atmeans
```

Conditional marginal effects   Number of obs   =   **15,262**
Model VCE   : **OIM**

Expression   : **Pr(has_ai), predict()**
dy/dx w.r.t. : **2.edu_cat 3.edu_cat 4.edu_cat 5.edu_cat 6.edu_cat experience_min_llm is_remote**
at   : 1.edu_cat   =   **.0215568** (mean)
               2.edu_cat   =   **.6126982** (mean)
               3.edu_cat   =   **.025095** (mean)
               4.edu_cat   =   **.0189359** (mean)
               5.edu_cat   =   **.3211244** (mean)
               6.edu_cat   =   **.0005897** (mean)
               experience~m   =   **4.520393** (mean)
               is_remote   =   **.2914428** (mean)

	Delta-method dy/dx	Std. Err.	z	P>|z|	[95% Conf.	Interval]
edu_cat						
bachelor	**.0960943**	**.0160304**	**5.99**	**0.000**	**.0646754**	**.1275132**
highschool	**.0310228**	**.0225205**	**1.38**	**0.168**	**-.0131166**	**.0751621**
master	**.1275784**	**.0290966**	**4.38**	**0.000**	**.0705501**	**.1846067**
missing	**.1640223**	**.0167551**	**9.79**	**0.000**	**.131183**	**.1968616**
phd	**.3567673**	**.1666906**	**2.14**	**0.032**	**.0300596**	**.6834749**
experience_min_llm	**.002914**	**.0012244**	**2.38**	**0.017**	**.0005142**	**.0053138**
is_remote	**.0651517**	**.0068062**	**9.57**	**0.000**	**.0518117**	**.0784916**

Note: dy/dx for factor levels is the discrete change from the base level.

```
290 .
291 .
292 . * ============================================================================
293 . * 7. ANALÝZA HARD SKILLS
294 . * ============================================================================
295 . * Rozbor nejčastějších technických dovedností
```

```
296 .
297 . display _n "==============================================================="

    ==============================================================

298 . display "7. ANALYZA HARD SKILLS"
    7. ANALYZA HARD SKILLS

299 . display "=============================================================="
    ==============================================================

300 .
301 . * Poznámka: hardskills je textový sloupec s čárkami oddělenými skills
302 . * Pro detailní analýzu je lepší použít Python k vytvoření dummy proměnných
303 . * Zde ukážeme základní exploraci
304 .
305 . * Počet unique skills na pozici (aproximace)
306 . gen skill_count = 1 + length(hardskills) - length(subinstr(hardskills, ",", "", .))

307 . replace skill_count = 0 if hardskills == ""
    (110 real changes made)

308 . label variable skill_count "Pocet pozadovanych hard skills"

309 .
310 . display _n "--- 7.1 Pocet skills na pozici ---"

    --- 7.1 Pocet skills na pozici ---

311 . summarize skill_count, detail

                  Pocet pozadovanych hard skills
    -------------------------------------------------------------
          Percentiles      Smallest
     1%            1              0
     5%            3              0
    10%            5              0       Obs              17,857
    25%            9              0       Sum of Wgt.      17,857

    50%           15                      Mean           16.78658
                            Largest       Std. Dev.      10.30584
    75%           23             72
    90%           31             72       Variance       106.2103
    95%           36             80       Skewness       .9649615
    99%           47             80       Kurtosis       4.325341

312 . tabstat skill_count, by(desc_tier_llm) statistics(count mean sd min max)

    Summary for variables: skill_count
         by categories of: desc_tier_llm
```

desc_tier_llm	N	mean	sd	min	max
ai_integration	2354	19.46432	10.3817	0	68
applied_ai	1307	20.62663	10.79929	1	80
core_ai	6	11.16667	6.431692	1	20
none	14190	15.99105	10.09171	0	80
Total	17857	16.78658	10.30584	0	80

```
313 .
314 . * T-test: Vyžadují AI pozice více skills?
315 . display _n "--- 7.2 T-test: Pocet skills AI vs non-AI ---"

    --- 7.2 T-test: Pocet skills AI vs non-AI ---

316 . ttest skill_count, by(has_ai)

    Two-sample t test with equal variances
    ------------------------------------------------------------------------------
       Group |     Obs        Mean    Std. Err.   Std. Dev.   [95% Conf. Interval]
    ---------+--------------------------------------------------------------------
           0 |  14,380    16.04652    .0842893    10.10769    15.88131    16.21174
           1 |   3,477    19.84728    .1789442    10.55164    19.49644    20.19813
    ---------+--------------------------------------------------------------------
    combined |  17,857    16.78658    .0771222    10.30584    16.63542    16.93775
    ---------+--------------------------------------------------------------------
        diff |           -3.800759      .19268               -4.178431   -3.423088
    ------------------------------------------------------------------------------
        diff = mean(0) - mean(1)                                      t = -19.7258
    Ho: diff = 0                                     degrees of freedom =    17855

        Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
     Pr(T < t) = 0.0000         Pr(|T| > |t|) = 0.0000          Pr(T > t) = 1.0000

317 .
318 .
319 . * ==============================================================================
320 . * 8. EXPORTY PRO TABULKY A GRAFY
321 . * ==============================================================================
322 . * Příprava dat pro publikovatelné tabulky
323 .
324 . display _n "=============================================================="

    ==============================================================

325 . display "8. EXPORTY"
    8. EXPORTY

326 . display "=============================================================="
    ==============================================================

327 .
328 . * --- 8.1 Summary statistics ---
329 . display _n "--- 8.1 Summary statistics pro tabulky ---"

    --- 8.1 Summary statistics pro tabulky ---

330 . summarize salary_mid experience_min_llm edu_cat skill_count has_ai is_remote

        Variable |        Obs        Mean    Std. Dev.       Min        Max
    -------------+---------------------------------------------------------
      salary_mid |     13,743    125240.4       42210      20000     450000
    experience~m |     15,262    4.520393     2.62528          0         35
         edu_cat |     17,857    3.115249    1.448891          1          6
     skill_count |     17,857    16.78658    10.30584          0         80
          has_ai |     17,857    .1947136    .3959911          0          1
    -------------+---------------------------------------------------------
       is_remote |     17,857    .2839223    .4509122          0          1
```

```
. 
. * Export do CSV pro další zpracování (grafy v Excelu/R/Python)
. export delimited desc_tier_llm salary_mid experience_min_llm edulevel_llm state sector using
  file ./output/summary_for_charts.csv saved

. 
. 
. * ============================================================================
. * 9. VIZUALIZACE
. * ============================================================================
. * Základní grafy pro diplomovou práci
. 
. display _n "============================================================"

  ============================================================

. display "9. VIZUALIZACE"
  9. VIZUALIZACE

. display "============================================================"
  ============================================================

. 
. * --- 9.1 Sloupcový graf: AI tier distribuce ---
. graph bar (count), over(desc_tier_llm) ///
  >     title("Distribuce AI pozadavku v IT pozicich") ///
  >     ytitle("Pocet pozic") ///
  >     bar(1, color(navy))

. graph export "$outdir/ai_tier_distribution.png", replace width(1200)
  (file ./output/ai_tier_distribution.png written in PNG format)

. 
. * --- 9.2 Box plot: Platy podle AI tier ---
. graph box salary_mid, over(desc_tier_llm) ///
  >     title("Rozlozeni platu podle AI pozadavku") ///
  >     ytitle("Rocni plat (USD)")

. graph export "$outdir/salary_by_ai_tier.png", replace width(1200)
  (file ./output/salary_by_ai_tier.png written in PNG format)

. 
. * --- 9.3 Histogram: Požadované zkušenosti ---
. histogram experience_min_llm if experience_min_llm < 15, ///
  >     by(has_ai, title("Pozadovane zkusenosti") note("")) ///
  >     xtitle("Roky zkusenosti") percent bin(15)

. graph export "$outdir/experience_histogram.png", replace width(1200)
  (file ./output/experience_histogram.png written in PNG format)

. 
. * --- 9.4 Vzdělání podle AI tier ---
. graph bar (count), over(edu_cat) over(has_ai) ///
  >     title("Vzdelavaci pozadavky: AI vs non-AI pozice") ///
  >     ytitle("Pocet pozic") ///
  >     legend(label(1 "non-AI") label(2 "AI pozice"))

. graph export "$outdir/education_by_ai.png", replace width(1200)
  (file ./output/education_by_ai.png written in PNG format)
```

```
.
.
. * ==========================================================================
. * 10. ZÁVĚR A ULOŽENÍ
. * ==========================================================================
.
. display _n "============================================================="

=============================================================

. display "ANALYZA DOKONCENA"
ANALYZA DOKONCENA

. display "============================================================="
=============================================================

. display "Vystupy ulozeny do: $outdir"
Vystupy ulozeny do: ./output

. display "Log soubor: $outdir/ai_skills_analysis_`time_string'.log"
Log soubor: ./output/ai_skills_analysis_24_Jan_2026_13-02-46.log

.
. * Ulož zpracovaný dataset pro další práci
. save "$outdir/ai_skills_processed.dta", replace
file ./output/ai_skills_processed.dta saved

.
. * Zavři log
. log close
      name:  <unnamed>
       log:  C:\Users\murcj\Projects\glassdoor-jobs-ai-skills\analysis\stata\output/ai_skills_a
  log type:  text
 closed on:  24 Jan 2026, 13:04:47
-----------------------------------------------------------------------------------------------

.
. * ==========================================================================
. * KONEC DO-FILU
. * ==========================================================================
.
end of do-file

.
.
```